

Miriam Shem

Software Engineer | AI Solutions Engineer | Digital Public Infrastructure

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PROFESSIONAL PROFILE

Software Engineer with professional experience across AI/ML systems, enterprise software, and government-facing integrations in East Africa. Built a production containerized AI pipeline (FastAPI, Celery, Redis, Docker, Whisper) for child protection organizations; contributed to an AI-powered KMS for a government energy organisation; engineered a multi-tenant corporate travel SaaS; and delivered enterprise billing and ERP integrations. Track record working with UNICEF, government ministries, and social-impact institutions. Advancing toward AI Product Leadership.

WORK EXPERIENCE

Software Engineer

2025 — Present

BITZ IT Consulting Ltd. · Nairobi, Kenya

Building and maintaining OpenCHS, an AI-powered open-source helpline and case management platform deployed across East Africa. Working across AI integration, backend engineering, system integrations with government and institutional stakeholders, production deployments, and product documentation.

Built production containerized AI pipeline (FastAPI, Celery, Redis, Docker, Whisper Large V3, spaCy) for audio → transcription → translation → NLP analysis, used by child protection organizations

Led OpenCHS integrations across Uganda MGLSD, Kenya CPIMS, WhatsApp, and SMS — deployed in 5 countries: Tanzania, Uganda, Kenya, Lesotho, Somalia

Worked directly with UNICEF Tanzania, UNICEF Kenya, Uganda MGLSD, C-Sema, and CPIMS Kenya as key stakeholders

Python FastAPI Django Java Spring Boot Vue 3 TypeScript Celery Redis
Docker Whisper NLP REST APIs PostgreSQL

Software Engineer (Intern → Full-time)

2023 — 2024

Ruel Software · Kenya

Joined as Software Support & Implementation Specialist Intern, supporting Ruel Enterprise ERP implementation and onboarding across 700+ client organizations in 8 sectors. Progressed to full-time Software Engineer, taking on enterprise integration and compliance work.

Progressed from Software Support & Implementation Specialist Intern to full-time Software Engineer within the same engagement

Led KRA eTIMS integration for Kenya Revenue Authority — tax compliance across 700+ enterprise clients

Delivered technical support, troubleshooting, and business process training across 8 industry sectors

ERP Systems

Enterprise Software

REST APIs

SQL

Technical Support

Integration

EDUCATION

Bachelor of Science — Computer Science

2019 – 2023

Kabarak University, Nakuru, Kenya

TECHNICAL SKILLS

LANGUAGES Python, JavaScript, TypeScript, Java, PHP, C#, R	BACKEND Django, FastAPI, Spring Boot, Node.js, Laravel	FRONTEND Vue 3, React, Next.js, Vite
AI / ML Whisper ASR, HuggingFace Transformers, NLLB (Translation), FLAN-T5 / Mistral, spaCy / NLP, Celery + Redis (ML queuing), TensorFlow / PyTorch, Data Science / Jupyter	APIS & INTEGRATIONS REST APIs, OAuth2 / JWT, RabbitMQ, Swagger / OpenAPI, SSL/TLS & API Security, Postman	TOOLS & DEVOPS Git / GitHub, Docker, Linux / Ubuntu / WSL, Label Studio, VS Code, Jupyter / Conda

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Digital Product Management: Modern Fundamentals — University of Virginia (Darden School of Business) · Coursera

In Progress

AI Product Management Specialization — Duke University · Coursera

Planned

KEY PROJECTS

- OpenCHS (East Africa):** AI-powered helpline & case management platform. Deployed in Tanzania, Uganda, Kenya, Lesotho. Clients include UNICEF Tanzania, UNICEF Kenya, Uganda MGLSD, C-Sema, CPIMS Kenya.
- Uganda Ministry of Gender Integration:** Django-based backend integration with the Uganda MGLSD gateway for child protection data exchange.
- Kenya CPIMS Integration:** Interoperability integration between OpenCHS and Kenya's national child protection registry (CPIMS).
- Tanzania AI Capabilities:** Speech-to-text (Whisper), translation (NLLB), summarization (FLAN-T5/Mistral) and classification for AI-enabled case processing.
- D-HEIGHT DPI Initiative:** Digital Public Infrastructure initiative covering housing, financial inclusion, WASH, and responsible AI. Confidential — research & design phase.
- KRA eTIMS Integration (Ruel Software):** Enterprise ERP integration with Kenya Revenue Authority for tax invoice compliance.